NBA Player Longevity Prediction – Feature Engineering

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("nba-players.csv")
print("Dataset shape:", df.shape)
df.head()
```

```
Dataset shape: (1340, 22)
```

	Unnamed: 0	name	gp	min	pts	fgm	fga	fg	3p_made	3pa	3p	ftm	fta	ft	oreb	dreb	reb	ast	stl	blk	tov	target
0	0	Brandon Ingram	36	27.4	7.4	2.6	7.6	34.7	0.5	2.1	25.0	1.6	2.3	69.9	0.7	3.4	4.1	1.9	0.4	0.4	1.3	
1	1	Andrew Harrison	35	26.9	7.2	2.0	6.7	29.6	0.7	2.8	23.5	2.6	3.4	76.5	0.5	2.0	2.4	3.7	1.1	0.5	1.6	
2	2	JaKarr Sampson	74	15.3	5.2	2.0	4.7	42.2	0.4	1.7	24.4	0.9	1.3	67.0	0.5	1.7	2.2	1.0	0.5	0.3	1.0	
3	3	Malik Sealy	58	11.6	5.7	2.3	5.5	42.6	0.1	0.5	22.6	0.9	1.3	68.9	1.0	0.9	1.9	0.8	0.6	0.1	1.0	

Target Variable and Class Balance

```
print(df['target_5yrs'].value_counts())
print()
print(df['target_5yrs'].value_counts(normalize=True))
```

```
target_5yrs
1    831
0    509
Name: count, dtype: int64

target_5yrs
1    0.620149
0    0.379851
Name: proportion, dtype: float64
```

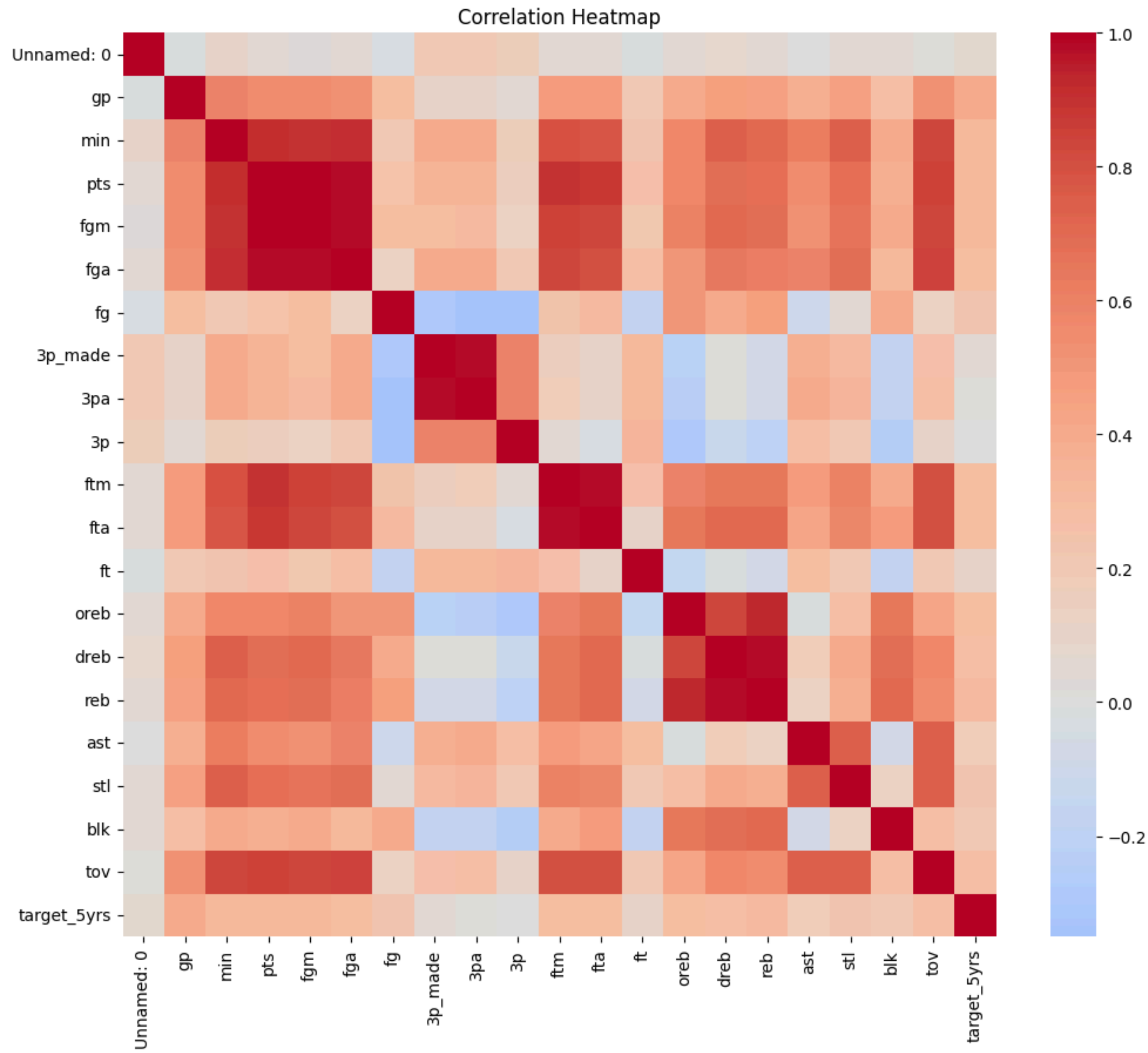
The target variable is `target_5yrs`. Examining class balance helps determine whether the dataset is biased.

```
target = df['target_5yrs']
drop_cols = [c for c in ['target_5yrs','name','Unnamed: 0'] if c in df.columns]
X = df.drop(columns=drop_cols)
print(X.columns.tolist())
```

```
['gp', 'min', 'pts', 'fgm', 'fga', 'fg', '3p_made', '3pa', '3p', 'ftm', 'fta', 'ft', 'oreb', 'dreb', 'reb', 'ast', 'stl', 'blk',
```

Correlation Heatmap and Multicollinearity

```
plt.figure(figsize=(12,10))
sns.heatmap(df.drop(columns=['name']).corr(), cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.show()
```




```
corr_matrix = X.corr().abs()

upper = corr_matrix.where(
    np.triu(np.ones(corr_matrix.shape), k=1).astype(bool)
)

high_corr_features = [
    col for col in upper.columns if any(upper[col] > 0.90)
]

print("Highly correlated features:", high_corr_features)
```

```
Highly correlated features: ['pts', 'fgm', 'fga', '3pa', 'fta', 'reb']
```

Features with correlation above 0.90 were considered redundant and removed to reduce multicollinearity.

Feature Engineering

```
X['points_per_minute'] = X['pts']/(X['min']+1e-6)

X['efficiency_rating'] = (
    X['pts'] + X['reb'] + X['ast'] + X['stl'] + X['blk']
)/(X['gp']+1e-6)
```

```
X = X.drop(columns=high_corr_features)

X.head()
```

	gp	min	fg	3p_made	3p	ftm	ft	oreb	dreb	ast	stl	blk	tov	points_per_minute	efficiency_rating
0	36	27.4	34.7	0.5	25.0	1.6	69.9	0.7	3.4	1.9	0.4	0.4	1.3	0.270073	0.394444
1	35	26.9	29.6	0.7	23.5	2.6	76.5	0.5	2.0	3.7	1.1	0.5	1.6	0.267658	0.425714
2	74	15.3	42.2	0.4	24.4	0.9	67.0	0.5	1.7	1.0	0.5	0.3	1.0	0.339869	0.124324
3	58	11.6	42.6	0.1	22.6	0.9	68.9	1.0	0.9	0.8	0.6	0.1	1.0	0.491379	0.156897
4	48	11.5	52.4	0.0	0.0	1.3	67.4	1.0	1.5	0.3	0.3	0.4	0.8	0.391304	0.166667

```
X = X.fillna(X.median(numeric_only=True))
print("Remaining missing values:", X.isnull().sum().sum())
```

```
Remaining missing values: 0
```

Stakeholder Summary

- `target_5yrs` was selected as the dependent variable.
- Non-predictive columns were removed.
- Correlation analysis identified redundant features.
- Points Per Minute and Efficiency Rating were engineered to measure player productivity.
- Median imputation handled missing values.
- The dataset is now ready for machine learning models.